5. Project Design Phase

Introduction

The **Project Design Phase** focuses on defining the structure and user interface of the **Smart Library Request Workflow in ServiceNow**. During this phase, the system's database tables, user interface behavior, and security controls are carefully designed to ensure smooth functionality and proper data management.

The design process includes creating structured tables to store book information, improving the user interface through UI policies, and implementing access control mechanisms to protect sensitive data. A well-designed system helps ensure that the application is easy to use, secure, and efficient.

Table Design

One of the most important tasks in the design phase is creating the **Book Table**, which stores all details related to the books available in the library. Proper table design ensures that the system can manage book information efficiently and maintain accurate records.

The **Book Table** includes several important fields that describe each book and its availability status.

Key Fields in the Book Table

- **Title** – The name of the book.
- **Author** – The person who wrote the book.
- **ISBN** – A unique identification number assigned to each book.
- **Category** – The subject or genre of the book.
- **Status** – Indicates whether the book is Available, Issued, or Lost.

Example Book Table Record

Book ID	Title	Author	ISBN	Status
B101	Data Structures	Mark Allen	978-1234567890	Available

When a student borrows the book, the **status automatically changes from "Available" to "Issued."**

Example Scenario

Suppose a book titled *"Artificial Intelligence Basics"* is added to the system. Initially, the status will be set to **Available**. If a student borrows the book, the system updates the status to **Issued**. If the book is lost or damaged, the librarian can change the status to **Lost**.

This table design allows the system to easily track book availability and maintain accurate records.

UI Enhancements

To improve usability and ensure proper data entry, **UI Policies** are implemented in the system. UI Policies control how fields behave on forms based on certain conditions.

In this project, UI policies are used to dynamically change form behavior depending on the status of a book. For example, the **Return Date field** becomes mandatory only when the book status is set to **Issued**.

This prevents users from entering unnecessary information and ensures that important details are recorded only when required.

Example UI Policy Logic

- If **Book Status = Issued**, then **Return Date = Mandatory**
- If **Book Status ≠ Issued**, then **Return Date = Not Mandatory**

Example Scenario

A librarian issues a book to a student. When the librarian selects the book status as **Issued**, the system automatically requires the **Return Date** field to be filled before saving the record.

If the book status is **Available**, the Return Date field will not be required.

Benefits of UI Enhancements

- Improves user experience by simplifying forms
- Prevents incomplete or incorrect data entry
- Ensures required information is collected at the right time
- Makes the application easier to use for librarians